

RAHUL MITRA

AI ENGINEER PROFILE

Singapore | mitrarahul2002@gmail.com | www.linkedin.com/in/rahulmitra-dev | github.com/Rah-Rah-Mitra | rahul-mitra.vercel.app

PROFILE SUMMARY

Early-career AI engineer focused on applied model systems, RAG, agentic workflows, NLP, computer vision, and AI-assisted creation pipelines. Built AI products across maritime, healthcare, sustainability, education, and geospatial monitoring.

EDUCATION

National University of Singapore - Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics - Relevant coursework: Artificial Intelligence; 3D Computer Vision; Software Engineering; Simulation; Statistics; Operations Research; Linear Algebra; Probability	Aug 2023 - Jul 2027
Anderson Serangoon Junior College - Singapore-Cambridge GCE A-Level	Feb 2019 - Dec 2020

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd - Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.	Jun 2026 - Present
Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd - Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules. - Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.	Jan 2026 - Jun 2026
Software Engineer (Citizen Developer), People's Association - Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore. - Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks. - Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.	Sep 2025 - Present

SELECTED PROJECTS AND COMPETITIONS

Maritime Deficiency Severity Forecasting - Led model training for a maritime inspection NLP system using a fine-tuned 109M-parameter BERT model, DNN layers, gradient clipping, and ASPIRE 2A supercomputing.	2025
AgeWellLah.AI - NUS HealthHack 2025 - Engineered an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, Flask, OAuth2-secured patient profiles, and PDPA-aware design choices.	2025
SmartExam - GovTech x NTUPC Product Hackathon - Built a multi-agent autonomous exam generation system with RAG, Next.js, source ingestion, question generation, and education workflow automation.	2025
EthosLens - LifeHack 2025 - Developed backend workflows for an AI research engine that scores brand ethics and sustainability practices using FastAPI, Python, LangChain, and automated evidence retrieval.	2025
Volt Pulse SG - SMU Hack For Cities Finalist - Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.	Jan 2026

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Python; BERT; DNN; RAG; AI Agents; OpenAI GPT-4; Gemini; Vector Search; SEALION; ASPIRE 2A; LangChain; Computer Vision; Model Evaluation.

- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre - Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.	Sep 2025 - Present
Astronomy Head and Robotics Member, STEM Inc. (ASRJC) - Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.	2019 - 2020